

CODE LAYOUT, READABILITY & REUSABILITY

Team ID: SWTID-2026-3238 | Project Name: PocketSmart AI | Track: AI-ML & Generative AI

1. Project Directory Structure

```
PocketSmart-AI/
├── app/
│   ├── config.py           # Pydantic BaseSettings loading .env configurations
│   ├── database.py         # SQLAlchemy engine, session maker, and Base model
│   ├── main.py             # FastAPI entrypoint, middleware, and router mounts
│   ├── schemas.py          # Pydantic request/response validation schemas
│   ├── models/             # SQLAlchemy ORM models (User, Recommendation)
│   ├── routers/            # APIRouter modules (auth.py, pages.py, planners.py)
│   ├── services/           # Business logic (auth, gemini, recommendations, providers)
│   ├── static/             # Assets (css/style.css, js/app.js, images/)
│   └── templates/          # Jinja2 templates (dashboard, home, party, jewelry, history)
├── tests/                  # Pytest verification suites (test_app.py)
├── requirements.txt        # Exact dependency manifest
├── .env.example            # Environment template for keys and DB paths
└── README.md              # Quickstart and project documentation
```

2. Modular Design & Clean Code Practices

- Adherence to PEP 8 naming standards, type annotations, and Pydantic schema validation.
- Complete decoupling between routing layer, business logic services, and persistence models.